

MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 7

Definition 1. A **principal ideal domain** or **PID** is an integral domain in which every ideal is principal.

(1) Decide if the following statements are true or false. Give a line or two in justification.

- (a) The quotient ring $k[x]/I$ with $I \leq k[x]$ an ideal is a field if and only if $I = (q(x))$, where $q(x) \in k[x]$ is an irreducible polynomial.

Solution. True. See Lecture 7.

- (b) There exists an injective k -linear transformation $T : V \rightarrow W$ between two k -vector spaces of dimension 4 that is *not* an isomorphism.

Solution. False. This would contradict rank-nullity, which says that any such linear transformation has to be surjective.

- (c) If V is a k -vector space admitting a k -subspace W such that $W \simeq k^2$, and $V/W \simeq k^3$, then $V \simeq k^5$.

Solution. True. If you already know that V is finite dimensional, this follows from Remark 1 in the solutions to HW 5. But actually finite dimensionality is for free, since any element of V can be written as a k -linear combination of a basis for W and any lift to V of a basis for V/W .

- (d) The matrix $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ is in rational canonical form.

Solution. False. The diagonal blocks of this matrix are of the form M_{x-1} and M_{x-2} and we do *not* have $x-1 \mid x-2$.

- (e) $\mathbb{Z}/n\mathbb{Z}$ is a PID.

Solution. False. It is not an integral domain.

(2) Show that we have the following natural isomorphisms of rings:

- (a) $\text{End}(\mathbb{Z}/n\mathbb{Z}) \simeq \mathbb{Z}/n\mathbb{Z}$ for all $n \in \mathbb{Z}$;

Solution. We will take the function

$$\text{End}(\mathbb{Z}/n\mathbb{Z}) \xrightarrow{f \mapsto f(1)} \mathbb{Z}/n\mathbb{Z}.$$

The fact that this is an isomorphism follows pretty much the same format as the case of $\text{End}(\mathbb{Z})$ considered in Lecture 19.

- (b) $\text{End}(\mathbb{Z}^r) \simeq M_{r \times r}(\mathbb{Z})$ where the right hand side is equipped with usual matrix addition and multiplication.

Solution. The function will take an endomorphism $f : \mathbb{Z}^r \rightarrow \mathbb{Z}^r$ to the square matrix A_f whose columns are given by

$$A_f = [f(e_1) \ \cdots \ f(e_r)].$$

This is a bijection because any endomorphism is determined completely and uniquely by where it sends the basis elements. The proof that it is a *ring* homomorphism goes along standard lines. In fact, matrix multiplication is essentially *rigged* to make this a ring homomorphism, since composition of homomorphisms is carried to multiplication of matrices. We used this last fact several times last semester in our proof of the existence of the Smith Normal Form.

(3) Use the existence of the rational canonical form to prove the following important fact: If $k \leq L$ is a field extension and $A, B \in M_n(k)$ are L -conjugate, then they are in fact k -conjugate.

Solution. The key observation is this: If $A' \in M_n(k)$ is the unique matrix in rational canonical form that is k -conjugate to A , then it is *also* the unique matrix in rational canonical form in $M_n(L)$ that is L -conjugate to A . The point is that being in rational canonical form does *not* depend on whether you think of a matrix as having coefficients in k or coefficients in L (why is this?).

Given this observation, we can now proceed as follows: Suppose that $B' \in M_n(k)$ is the unique matrix in rational canonical form that is k -conjugate to B . If we knew that $B' = A'$ then we can conclude that B and A are k -conjugate (by transitivity of this property). However, we *do* know that B and A are L -conjugate, and this tells us that the rational canonical forms B' and A' *must* be equal in $M_n(L)$ (why?). But then they must already be equal in $M_n(k)$ (why?).

For any $A \in M_n(k)$, denote the k -pair (k^n, T_A) by V_A . For any monic non-constant polynomial $q(x)$, write V_q for the k -pair $(k[x]/(q(x)), T)$, where T is given by multiplication-by- x .

Definition 2. A matrix $A \in M_n(k)$ is **diagonalizable** if it is k -conjugate to a diagonal matrix.

(4) Show that the following are equivalent for a matrix $A \in M_n(k)$:

- (a) A is diagonalizable over k .
- (b) The k -pair V_A is isomorphic to $V_{x-\lambda_1} \times \cdots \times V_{x-\lambda_n}$ for some $\lambda_1, \dots, \lambda_n \in k$.

Solution. We will use the following interpretation of Problem 2 from Homework 6: For two matrices $A, B \in M_n(k)$, we have $V_A \simeq V_B$ if and only if A and B are k -conjugate.

The main point is that $V_{x-\lambda}$ is isomorphic to $(k, [\lambda]) = V_{[\lambda]}$, where we think of $[\lambda]$ as a 1×1 -matrix: Just use the basis $1 + (x - \lambda)$ for $k[x]/(x - \lambda)$ and note that multiplication by x carries it to $\lambda + (x - \lambda)$. Then by Problem 5 of Homework 6, we see

$$V_{x-\lambda_1} \times \cdots \times V_{x-\lambda_n} \simeq V_B$$

$$\text{where } B = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ 0 & \ddots & 0 \\ 0 & \cdots & \lambda_n \end{bmatrix}.$$

So both (1) and (2) are equivalent to saying that $V_A \simeq V_B$ where B is the above diagonal matrix for some choice of $\lambda_1, \dots, \lambda_n \in k$.

If $\lambda \in k$ and $n \in \mathbb{Z}_{\geq 1}$, the **Jordan block matrix** $J_\lambda(n) \in M_n(k)$ is the matrix

$$J_\lambda(n) = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{bmatrix} \in M_n(k)$$

with λ s along the diagonal, and 1s just above the diagonal.

(5) Suppose that $q(x) = (x - \lambda)^n \in k[x]$. Show that the k -pair V_q is isomorphic to $V_{J_\lambda(n)}$.

Hint: You'll need to pick the right basis for V_q !

Solution. Pick the basis $\mathcal{B} = \{v_1, \dots, v_n\}$, where $v_i = (x - \lambda)^{n-i} + (x - \lambda)^n$, and note that we have

$$x \cdot v_i = (x - \lambda + \lambda) \cdot (x - \lambda)^{n-i} = (x - \lambda)^{n-i+1} + \lambda \cdot (x - \lambda)^{n-i} = v_{i-1} + \lambda \cdot v_i.$$

Here, if $i = 1$, then we take $v_0 = 0$, since we would get $(x - \lambda)^{n-1+1} = (x - \lambda)^n = 0 \in k[x]/((x - \lambda)^n)$.

Unwinding definitions now shows that the $\mathcal{B}$ -matrix for the multiplication by x map on $k[x]/((x - \lambda)^n)$ is exactly $J_\lambda(n)$. More precisely, $J_\lambda(n)$ is the matrix whose i -th column is $e_{i-1} + \lambda \cdot e_i$, where once again we will take $e_0 = 0$.

Definition 3. The **ring of Gaussian integers** $\mathbb{Z}[i]$ is the subset of $\mathbb{C}$ consisting of elements $a + bi$ with $a, b \in \mathbb{Z}$. This is in fact a sub-ring of $\mathbb{C}$.

- (6) Show that, for $\alpha, \beta \in \mathbb{Z}[i]$ with $\alpha \neq 0, \pm 1, \pm i$, there exist $\kappa, \rho \in \mathbb{Z}[i]$ with $|\rho| < |\alpha|$ such that $\beta = \kappa\alpha + \rho$. Use this to verify that $\mathbb{Z}[i]$ is a PID.¹

Hint: Looking at β/α , you just have to show that every element of $\mathbb{C}$ has distance less than 1 to the closest point with integer co-ordinates.

Solution. As the hint says, we consider $\beta/\alpha \in \mathbb{C}$. By looking at the grid with integer coordinates, we see that there is actually an element $\kappa \in \mathbb{Z}[i]$ with

$$\left| \frac{\beta}{\alpha} - \kappa \right| \leq \frac{1}{\sqrt{2}}.$$

That is, any point inside a unit square is at most $\frac{1}{\sqrt{2}}$ away from one of the corners.

Now, this shows that $\rho = \alpha \cdot \left(\frac{\beta}{\alpha} - \kappa \right) = \beta - \alpha\kappa$ satisfies $|\rho| \leq \frac{1}{\sqrt{2}}|\alpha| < |\alpha|$.

We can now follow the proofs used to show that $k[x]$ and $\mathbb{Z}$ are PIDs: Given a non-zero ideal $I \leq \mathbb{Z}[i]$ choose $0 \neq \alpha \in I$ such that $|\alpha|^2$ is the smallest possible. This makes sense because $|\alpha|^2$ is always a positive integer, so there has to be a smallest possible value for non-zero elements of I .

I claim that $(\alpha) = I$. Indeed, if $\beta \in I$ and we write $\beta = \kappa\alpha + \rho$ as above, then $\rho = \beta - \kappa\alpha \in I$ (why?) and we have $|\rho| < |\alpha|$. By our choice of α , this can only happen if $|\rho| = 0$, which can only happen if $\rho = 0$. In other words, $\beta = \kappa\alpha \in (\alpha)$. The other inclusion $(\alpha) \leq I$ is trivial.

Remark 1. The proof applies quite generally in the following situation. Suppose that R is an integral domain and that we have a function $N : R \rightarrow \mathbb{Z}_{\geq 0}$ with the following properties:

- (a) $N(r) = 0$ if and only if $r = 0$;
 - (b) Given non-zero elements $x, y \in R$, we can find $q, r \in R$ with $N(r) < N(x)$ such that $y = qx + r$.
- Such a function is called a **Euclidean norm**. If R is equipped with such a norm, then the argument above shows that R has to be a PID.

¹This is the complex absolute value here given by $|a + ib| = \sqrt{a^2 + b^2}$